Lecture no. 12 :-

Spam Detection Maths - Bayes Rule Real World

Applying Bayes Rule to Spam Detection :-

- "You have inherited a million dollars".
- "There will be a meeting at noon".
- Assumption : We have a data set of spam emails.
- Need to find whether a piece of text is spam.

- Let's first consider a single word.

"how frequently does this word appear in spam?"

"how much spam is there in the world?"

$$\underbrace{P(spam|w)}_{\text{"Given that this word appears, how likely is it that the message is spam?"}} = \frac{P(w|spam)\ P(spam)}{\underbrace{P(w)}_{\text{from dataset "How frequent is this word?"}}}$$

$$P(spam) = \frac{\text{No. of spam messages}}{\text{No. of all messages}}$$

$$P(w \mid spam) = \frac{\text{no. of times this word appears in spam}}{\text{no. of spam messages}}$$

$$P(w) = \frac{\text{No. of times this word appears}}{\text{No. of total messages}}$$

Now, do this for all words

$$P(spam \mid words) = P(spam \mid w_1) * P(spam \mid w_2) * \dots * P(spam \mid w_n)$$

$$\boxed{P(spam \mid words) = \prod_{i=1}^{|words|} P(spam \mid w_i)}$$

$\because$ Σ is for submittion

So Π is for product

(product of all items in eq)